

# Database Design & SQL

## Introduction

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# Learning Objectives

- Why to use database
- The difference between data and information
- What is a database?
- Outline the main components of the database system
- Describe the main functions of a database management system (DBMS)
- The Characteristics of Databases
- Relational Databases

# Why Databases?

## A Day in Susan's Life

See how many databases she interacts with each day

*In the morning, Susan goes grocery shopping*



Where are the product data stored?  
Is the product quantity in stock updated at checkout?  
Does she pay with a credit card?



*Later, she picks up her prescription at the pharmacy*



Where is the pharmacy inventory data stored?  
What data about each product will be in the inventory data?  
What data is kept about each customer and where is it stored?



*In the afternoon, she orders some items online*



Where are the product and stock data stored?  
Where does the system get the data to generate product "recommendations" to the customer?



*At night, she plans for a trip and buys airline tickets and hotel reservations online*



Where does the online travel website get the airline and hotel data from?  
What customer data would be kept by the website?  
Where would the customer data be stored?



*Before going to bed, Susan checks her social media accounts*



What particular customer data is kept by each website?  
Where is the data about the friends and groups stored?  
Where are the "likes" stored and what would they be used for?



**Figure 1.1** The Pervasive Nature of Databases

# Data versus Information (1 of 2)

**Data** consists of raw facts

The facts have not yet processed to reveal meaning to the end user

**Information** is the result of processing raw data to reveal the meaning of data  
To reveal meaning.

- Data is the foundation of information.
- Accurate, relevant, and timely information is the key to good decision making.
- **Data management** is a discipline that focuses on the proper generation, storage, and retrieval of data.

# Data versus Information (2 of 2)

a) Data entry screen

b) Raw data

| ID | Last Name | First Name | Dept      | Col  | Office         | E-mail              | Rank                | Next Year Degree |
|----|-----------|------------|-----------|------|----------------|---------------------|---------------------|------------------|
| 1  | Wright    | A          | George    | MGMT | 1025           | gwright@mtsu.edu    | Professor           | 2003 Ph.D.       |
| 2  | Adams     | J          | John      | FIN  | 1025           | jadams@mtsu.edu     | Professor           | 1995 Ph.D.       |
| 3  | Jefferson | L          | Thomas    | ECOM | 1025           | ljefferson@mtsu.edu | Instructor          | 2002 M.F.A.      |
| 4  | Madison   | B          | James     | FIN  | 1026           | jmadison@mtsu.edu   | Associate Professor | 1998 Ph.D.       |
| 5  | Moore     | R          | James     | ACCT | 1021           | jmoore@mtsu.edu     | Associate Professor | 1995 Ph.D.       |
| 6  | Adams     | G          | John      | ACCT | 1025           | jadams@mtsu.edu     | Associate Professor | 1995 Ph.D.       |
| 7  | Jackson   | C          | Andrew    | ECOM | 1021           | ajackson@mtsu.edu   | Associate Professor | 1999 Ph.D.       |
| 8  | Wright    | E          | Mark      | FIN  | 1026           | ewright@mtsu.edu    | Professor           | 1995 Ph.D.       |
| 9  | Harrison  | R          | William   | MGMT | 1011           | wharrison@mtsu.edu  | Professor           | 1994 Ph.D.       |
| 10 | Tyler     | M          | John      | MGMT | 1025           | mtyler@mtsu.edu     | Associate Professor | 2000 Ed.D.       |
| 11 | Park      | Gregg      | MGMT      | 1011 | gpark@mtsu.edu | Associate Professor | 2002 Ph.D.          |                  |
| 12 | Taylor    | G          | Zachary   | ACCT | 1021           | gtaylor@mtsu.edu    | Associate Professor | 1998 Ph.D.       |
| 13 | Finney    | A          | Michael   | JOB  | 1021           | afinney@mtsu.edu    | Professor           | 1992 Ph.D.       |
| 14 | Pierce    | A          | Frederick | MGMT | 1025           | fpierce@mtsu.edu    | Instructor          | 2005 M.F.A.      |
| 15 | Farrell   | F          | James     | MGMT | 1016           | jfarrell@mtsu.edu   | Associate Professor | 1998 D.S.A.      |
| 16 | Lewis     | W          | Larry     | MGMT | 1020           | lwlewis@mtsu.edu    | Associate Professor | 1998 Ph.D.       |
| 17 | Johnson   | A          | Andrew    | ECOM | 1021           | ajohnson@mtsu.edu   | Professor           | 1997 Ph.D.       |
| 18 | Gunn      | K          | Katie     | MGMT | 1020           | kgunn@mtsu.edu      | Associate Professor | 1999 D.S.A.      |
| 19 | McKinnon  | S          | Shirley   | ACCT | 1025           | smckinnon@mtsu.edu  | Professor           | 1992 Ph.D.       |
| 20 | Graham    | E          | David     | ACCT | 1025           | dgraham@mtsu.edu    | Associate Professor | 2001 Ph.D.       |
| 21 | Decker    | S          | Emily     | ACCT | 1025           | edecker@mtsu.edu    | Associate Professor | 2001 D.D.        |
| 22 | Calverton | S          | Robert    | ACCT | 1021           | rcalverton@mtsu.edu | Associate Professor | 1997 Ph.D.       |
| 23 | Harrison  | N          | Patricia  | BUS  | 1020           | nharrison@mtsu.edu  | Associate Professor | 2001 D.D.        |
| 24 | McKinnon  | B          | Frederick | MGMT | 1025           | fmckinnon@mtsu.edu  | Adjunct             | 1996 M.F.        |
| 25 | Finney    | F          | Michael   | MGMT | 1024           | mfinney@mtsu.edu    | Associate Professor | 2002 Ph.D.       |
| 26 | Moore     | L          | Laura     | ECOM | 1025           | lmoore@mtsu.edu     | Professor           | 1992 Ph.D.       |
| 27 | Finney    | A          | Whitney   | MGMT | 1024           | wfinney@mtsu.edu    | Professor           | 1994 Ed.S.       |
| 28 | McKinnon  | G          | Carol     | ECOM | 1025           | cmckinnon@mtsu.edu  | Professor           | 1975 Ph.D.       |
| 29 | Harrison  | L          | Lee       | MGMT | 1025           | lharrison@mtsu.edu  | Adjunct             | 1975 M.F.A.      |
| 30 | Truman    | D          | Darryl    | ACCT | 1025           | dtruman@mtsu.edu    | Professor           | 1977 Ed.D.       |
| 31 | Johnson   | A          | Robert    | ECOM | 1025           | ajohnson@mtsu.edu   | Professor           | 2000 Ph.D.       |

c) Information in summary format

| Rank                | COUNT | %INF.  | TOT.COL | %COL. TOT. | %COL. FAC. |
|---------------------|-------|--------|---------|------------|------------|
| Adjunct             | 5     | 20.00% | 23      | 21.74%     | 3.23%      |
| Assistant Professor | 2     | 8.00%  | 28      | 7.14%      | 1.31%      |
| Associate Professor | 9     | 36.00% | 37      | 24.32%     | 5.68%      |
| Instructor          | 2     | 8.00%  | 16      | 11.11%     | 1.31%      |
| Professor           | 7     | 28.00% | 47      | 14.89%     | 4.58%      |

d) Information in graphical format

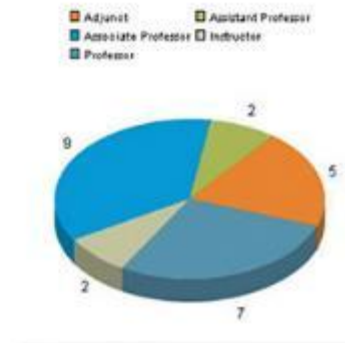


Figure 1.2 Transforming Raw Data into Information

# Database

A **database** is an organized collection of related data:

- Represent some aspect of the real world,
- Stored electronically in a computer system.

A **database** is usually controlled by a **database management system (DBMS)**. Together, the data and the DBMS, along with the applications that are associated with them, are referred to as a **database system**.

Example of a large commercial database system: **Amazon.ca**



# Examples of Databases use

Have you ever filled out a form online?

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Your info was stored in a database.

Have you ever browsed Amazon or any online retailer?

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Every item you are shown is being retrieved from a database.

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Every sale that is made is recorded to a database

Access to your medical records

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The results of medical tests and your prescriptions are stored in a database

# Database Management System (DBMS)

- A **DBMS** is a software that manages the databases and controls access to the data stored in the database.
- A DBMS serves as an interface between the database and its users, allowing users to retrieve, update, and manage how the information is organized and optimized.
- Some examples of popular database software or DBMSs include
  - MySQL
  - Microsoft Access
  - Microsoft SQL Server
  - Oracle Database
  - PostgreSQL



# End-User-Data and Metadata

A database stores a collection of the following:

- **End-user data** : raw facts of interest to the end user.
- **Metadata**, or data about data, through which the end-user data is integrated and managed.

# Metadata

- Metadata describes the data characteristics and the set of relationships that links the data found within the database
- Database definition or descriptive information stored by the DBMS in the form of a database catalog or dictionary

**DATA**

| employee_id | first_name | last_name  | nin           | dept_id |
|-------------|------------|------------|---------------|---------|
| 44          | Simon      | Martinez   | HH 45 09 73 D | 1       |
| 45          | Thomas     | Goldstein  | SA 75 35 42 B | 2       |
| 46          | Eugene     | Comelsen   | NE 22 63 82   | 2       |
| 47          | Andrew     | Petculescu | XY 29 87 61 A | 1       |
| 48          | Ruth       | Stadick    | MA 12 89 36 A | 15      |
| 49          | Barry      | Scardelis  | AT 20 73 18   | 2       |
| 50          | Sidney     | Hunter     | HW 12 94 21 C | 6       |
| 51          | Jeffrey    | Evans      | LX 13 26 39 B | 6       |
| 52          | Doris      | Bemdt      | YA 49 88 11 A | 3       |
| 53          | Diane      | Eaton      | BE 08 74 68 A | 1       |

**DATA DICTIONARY (METADATA)**

| Column                | Data Type    | Description                               |
|-----------------------|--------------|-------------------------------------------|
| employee_id           | int          | Primary key of a table                    |
| first_name            | nvarchar(50) | Employee first name                       |
| last_name             | nvarchar(50) | Employee last name                        |
| nin                   | nvarchar(15) | National Identification Number            |
| position              | nvarchar(50) | Current position title, e.g. Secretary    |
| dept_id               | int          | Employee department. Ref: Departments     |
| gender                | char(1)      | M = Male, F = Female, Null = unknown      |
| employment_start_date | date         | Start date of employment in organization. |
| employment_end_date   | date         | Employment end date.                      |

# Database System

Users  
Programmers

DBMS

Database

Metadata

PRODUCTS

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|  |  |  |  |
|  |  |  |  |

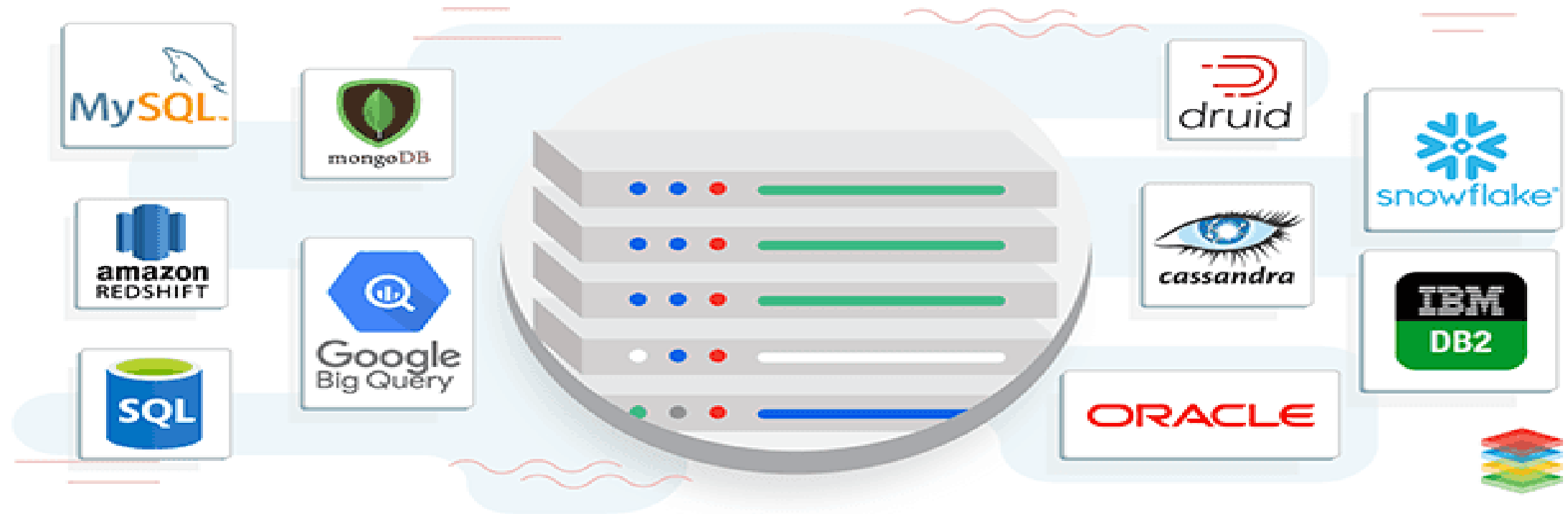
CUSTOMERS

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|  |  |  |  |
|  |  |  |  |

ORDERS

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|  |  |  |  |
|  |  |  |  |

# List of Popular Databases



# Databases types

## Relational databases

- ✓ Items are organized as a set of **tables** with **columns** and **rows**.
- ✓ Provides the most efficient and flexible way to access structured information.

## Object-oriented databases

- ✓ Information in an object-oriented database is represented in the form of objects, as in object-oriented programming.

## Distributed databases.

- ✓ Consists of two or more files located in different sites.
- ✓ The database may be stored on multiple computers, located in the same physical location, or scattered over different networks.

## Cloud databases

- ✓ Resides cloud database is a on a private, public, or hybrid cloud computing platform.

## NoSQL databases

- ✓ New generation of DBMS that is not based on the traditional relational database model.
- ✓ Designed to handle the unprecedented volume of data, variety of data types and structures, and velocity of data operations that are characteristic of these new business requirements

# The Characteristics of Databases

- Data is stored in **tables**, which have **rows** and **columns** like a spreadsheet
- Each row in a table stores data about an occurrence or **instance** of the thing of interest
- **Tuple**: Rows or records
- **Attribute**: Columns or fields
- **Primary Key (PK)**: It is a unique identifier, such as a driver license number, telephone number (including area code), or vehicle identification number (VIN)

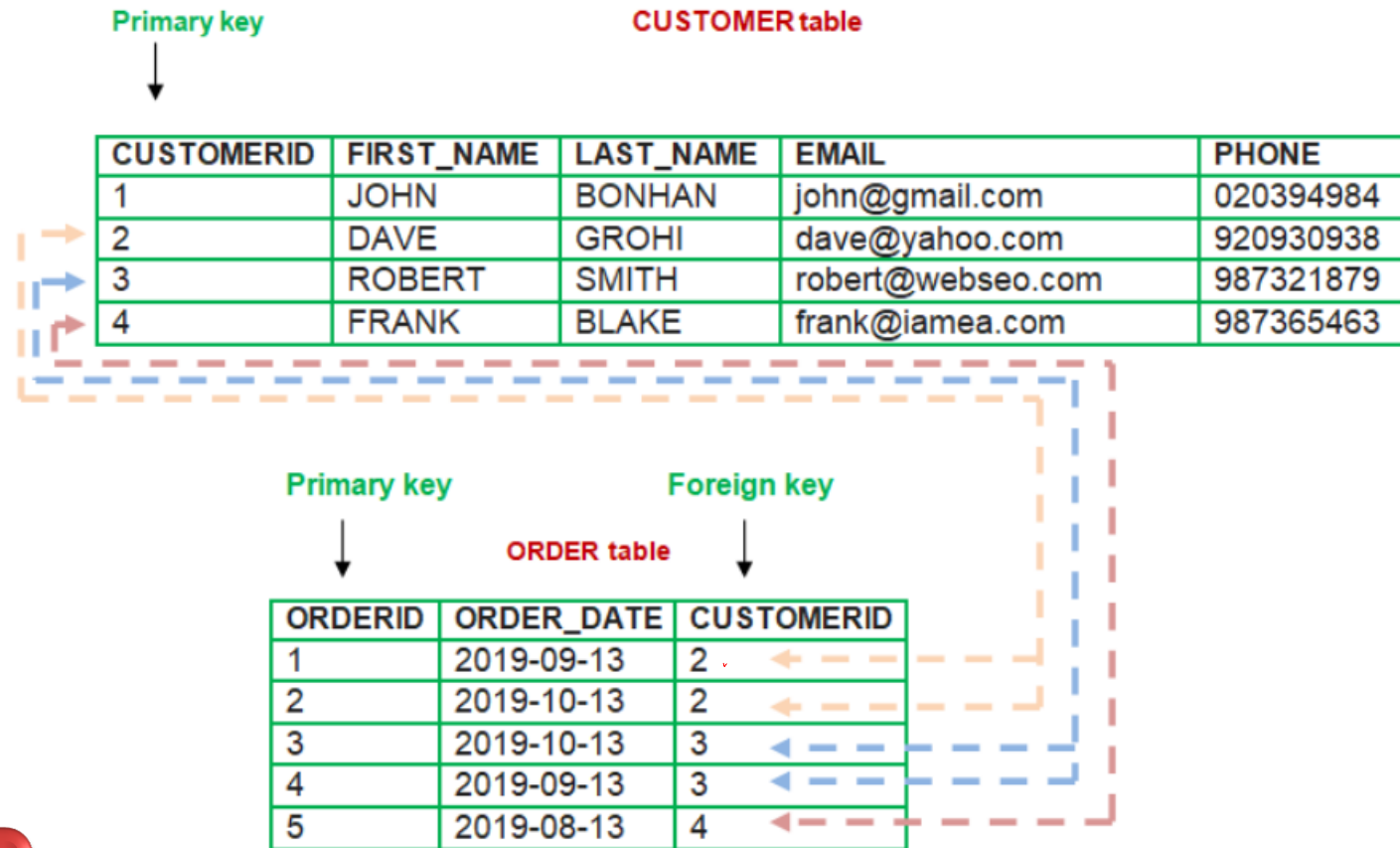
The diagram illustrates a database table with four columns: CustomerID, FirstName, LastName, and Birthdate. The table contains four rows of data. Annotations with red arrows and boxes identify key components: 'Column (attribute)' points to the column headers; 'Table (relation)' points to the entire table structure; 'Row (tuple)' points to the second row; 'Primary key' points to the CustomerID column; and 'Data value' points to the value 'Green' in the LastName column of the fourth row.

| CustomerID | FirstName | LastName        | Birthdate          |
|------------|-----------|-----------------|--------------------|
| XY001      | John      | Doe             | April 18, 1929     |
| BR092      | Mary      | Green           | March 4, 1980      |
| PD500      | Francesca | de la Gillebert | September 12, 1959 |
| WI308      | John      | Green           | March 4, 1980      |

# Relational Databases (proposed by Codd in 1970)

- **Primary Key**: It is a unique identifier, such as a driver license number...
- **Foreign Key**: is a column in a relational database table that exists as the primary key in another table and provides a **link** between data in these two tables.

A primary key-foreign key relationship



From now on when we say database, we refer to relational database.

DBMS → RDBMS